

Assessing Stability of Local Explanations: A Feature Importance Metric for Temporal Data

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Abstract. Feature importance plays a critical role in the explainability of machine learning models. In the context of temporal data, where more data becomes available over time, feature importance may change to varying degrees. However, these changes remain underexplored in the literature. This motivates us to quantify fluctuations in feature importance to improve the interpretability of models trained on temporal data. In this paper, we propose a novel method to measure the stability of local explanations by tracking changes in feature importance over time. The proposed approach helps to assess the evolving relevance of features as new data arrives. By incorporating Exponentially Weighted Moving Average and performance-based weighting, our metric captures both feature fluctuations and variations in model performance, thereby improving interpretability. The considered exponential smoothing is able to emphasize the most recent data, while the performance-based weighting accounts for differences in prediction accuracy. To evaluate our approach, we conducted experiments on benchmark time series datasets and real-world medical data from patients diagnosed with bipolar disorder. We simulated scenarios where the amount of available patient data changes over time, various types of prior distributions and analyzed how feature importance evolves in sequential models. Our approach allows for the re-computation of feature contributions, providing insight into the stability of feature attributions over time.

Keywords: Explainable AI · Feature Importance · Time Series Analysis · Shapley Values

1 Introduction

Explainable AI (XAI) has made significant progress [12], but several challenges remain, as highlighted in earlier works such as [11]. Further research is needed to address these issues, especially in emerging applications. This need is even more pressing given the increasing demand for trustworthy AI systems, as highlighted by regulatory efforts such as the Ethics Guidelines for Trustworthy AI [8] or the European Union’s AI Act [2], where explainability is identified as a key requirement. While explainability has been extensively studied in static machine learning models, their applications to temporal data present unique challenges. Although the research on explainability for time series models is dynamically growing, it still remains an underexplored area [23, 4, 21]. The dynamic nature

of temporal data complicates the measurement of feature importance, which is a fundamental aspect of model interpretability [20]. Understanding these fluctuations is critical since feature importance can vary significantly over time. The majority of the common methods allow for the measurement of feature importance at a given point in time [15] or over an interval [23], they may overlook how this importance evolves. Capturing a feature’s stability—its consistent contribution to a model’s output over different time periods—can provide valuable insights into the reliability of the learned relationships.

Feature stability has long been recognized as a critical aspect of machine learning, particularly in feature selection, where the goal is to ensure that selected features remain consistent across different datasets or model variations. Previous research, such as [5], has explored the relationship between feature selection methods and their stability, highlighting its importance in building robust and interpretable models. However, much of this work has been conducted in the context of static datasets, with limited attention given to temporal data.

To address this gap, we propose a novel approach that quantifies fluctuations in feature importance over time accounting for the performance and changes of predictions as more data becomes available, thereby improving the interpretability of temporal machine learning models. We validate our approach using a diverse set of datasets, including a case study in mental health monitoring and benchmark datasets from the UCI Machine Learning Repository [3, 19]. Our results demonstrate that the proposed method provides valuable insights into the dynamics of feature importance over time, considering both the temporal nature of the data and model performance. The proposed metric captures the magnitude and direction of fluctuations in feature values, which, when incorporated into a broader framework, can offer a more comprehensive understanding of feature behavior and improve feature selection.

The structure of the paper is as follows. In the next section, we present a brief description of related works. Section 3 presents the basic definitions related to the Shapley values. This is followed by a presentation of the proposed approach in Section 4. The experimental results using real-life medical data, simulated data and benchmark datasets are presented in Section 5. Finally, Section 6 outlines the conclusions and discusses directions for future work.

2 Related Work

One of the ways to explain a machine learning model’s behavior is to analyze how a given variable contributes to the model’s output. A central concept in such analyses is feature importance [16]. Numerous methods exist for measuring feature importance, each offering different perspectives on a model’s decision-making process [9, 20]. These methods can be broadly categorized into model-specific and model-agnostic approaches. Some machine learning models have built-in mechanisms for estimating feature importance. For example, Random Forests and Gradient Boosting Machines use metrics such as Gini importance and gain-based importance to quantify how much each feature contributes to

the model’s predictive performance [10]. Similarly, linear models determine feature importance based on their coefficient values, where larger absolute values indicate stronger influence on predictions [13]. Despite their advantages, model-specific importance measures have inherent limitations. They are tightly linked to the model’s structure and assumptions, which can limit their interpretability when applied across different architectures.

Model-agnostic approaches offer greater flexibility by evaluating feature importance independently of the underlying model. Permutation importance and Shapley values are two widely used methods in this category. The first method, estimates a feature’s relevance by randomly shuffling its values and observing the impact on model performance [9]. While computationally efficient, this method is sensitive to feature correlations, which can distort the true importance of individual variables [26]. Shapley values, derived from cooperative game theory, provide a theoretically grounded approach to feature attribution. They quantify a feature’s importance based on its marginal contribution to all possible feature subsets, ensuring a fair distribution of attributions [24]. However, exact computation is often infeasible for high-dimensional datasets. To address this, the SHapley Additive Explanations (SHAP) framework was introduced, offering efficient approximations of Shapley values while providing both local and global explanations of model behavior [17].

While feature importance has been extensively studied for static datasets, applying these methods to temporal data presents additional challenges. Traditional feature importance measures often fail to capture shifts in feature relevance over time, treating each time step independently and overlooking the stability or variability of feature importance across different time points [27]. To address these challenges, new methodologies are required that account for the temporal dynamics in feature attributions, enabling a more robust understanding of how model behavior evolves over time [15]. Recent approaches attempt to integrate temporal dependencies into feature importance estimates, ensuring that the analysis remains faithful to the underlying data structure.

The evaluation of model explanations has emerged as a pivotal research focus in recent years, as evidenced by the growing body of literature addressing this subject [22, 18]. These studies underscore the imperative for evaluating explanations from diverse vantage points, emphasising that disparate evaluation metrics encompass discrete facets of explainability. A thorough analysis of these aspects enhances our understanding of widely used feature importance measures, such as Shapley values. The incorporation of objective evaluation metrics into explanation methods has the potential to offer more profound insights into model behavior, which may result in enhanced interpretability and, in certain instances, improved predictive performance.

In this work, we propose a novel metric designed to evaluate the stability of local explanations by capturing the temporal nature of data. The method reflects the scenario where more data becomes available over time, potentially leading to improved model performance. To validate whether additional data

truly enhances the model output, we introduce performance-based weighting, which reduces the impact of poorly performing models on feature importance.

3 Shapley Values for Feature Importance

We now recall basic definitions related to the Shapley values, which are a game-theoretic approach aiming to ensure the attribution of feature importance is fair and consistent. Let $D = \{1, 2, \dots, N\}$ be the set of all players, and let $\nu : 2^N \rightarrow \mathbb{R}$ be a utility function that assigns a real number $\nu(S)$ to each possible coalition S , representing the utility of that coalition, e.g., the profit it can generate. The Shapley value $s(\nu, i)$ originally introduced in [24] for player i with respect to the utility function ν is defined as the sum of their marginal contributions to coalition S over all permutations of the player set in which player i joins the coalition S , where $S \subseteq D \setminus \{i\}$:

$$s(\nu, i) = \frac{1}{|D|} \sum_{S \subseteq D \setminus \{i\}} \frac{1}{\binom{|D|-1}{|S|}} [\nu(S \cup \{i\}) - \nu(S)]. \quad (1)$$

For a fixed utility function ν , the Shapley value for the i -th player can be denoted as $s(\nu, i) = s_i$.

In practice, computing the Shapley values directly for each feature subset is computationally expensive, especially for datasets with a large number of features. Due to the exponentially growing number of possible feature subsets 2^N , where N is the number of features, computing the exact Shapley value for high-dimensional datasets is almost infeasible.

The common The Shapley Additive Explanations (SHAP) framework [17] has facilitated the widespread applicability of Shapley values in practice by providing computationally feasible approximations. Tree-based models, in particular, have been shown to benefit from SHAP’s efficiency gains, enabling the utilization of Shapley values in high-dimensional, complex models.

For a given model f and an input x , the SHAP value $\phi_i(x)$ for feature i is given by:

$$\phi_i(x) = \sum_{S \subseteq D \setminus \{i\}} \frac{|S|!(|D| - |S| - 1)!}{|D|!} [f(S \cup \{x_i\}) - f(S)] \quad (2)$$

where $f(S \cup \{x_i\})$ is the prediction of the model when feature i is included in the set S and $f(S)$ is the prediction of the model when feature i is excluded from the set S . This formalism satisfies properties such as efficiency, symmetry, and additivity. This method offers clear insights into the contribution of individual features to the model’s decision-making process.

4 Assessing Stability of Local Explanations for Temporal Data

In this section, we introduce a novel metric designed to quantify temporal variations in feature importance for multivariate time-series data. Our approach lever-

ages the Exponentially Weighted Moving Average (EWMA) and a performance-based weighting scheme to assess feature stability, thereby enhancing model interpretability over time. By incorporating both temporal dynamics and model performance, our proposed metric provides a feature-specific measure of stability, ensuring the recent data is weighted appropriately while mitigating the influence of poorly performing models.

Let $\mathbf{X} \in \mathbb{R}^{K \times T}$ denote a sample of multivariate time series with K features and T time points. Let also $\mathbf{x}_t := \mathbf{X}_{:,t} \in \mathbb{R}^K$ be the set of all feature observations at a time point $1 \leq t \leq T$, where $T > 1$. The feature importance scores are computed based on the realizations $\mathbf{x}_t$ of the time series. Let $F_t^{(j)}$ represent importance score of the feature j at the time point t , derived from a model-specific feature attribution method $F^{(j)} = \{F_1^{(j)}, F_2^{(j)}, \dots, F_T^{(j)}\}$.

The stability metric is defined as:

$$SM(j) = 1 - \frac{1}{n-1} \sum_{t=2}^n g(w_t) \frac{\left(F_t^{(j)} - \hat{F}_t^{(j)}\right)^2}{|F_t^{(j)}| + |\hat{F}_t^{(j)}|}, \quad (3)$$

where:

- $g(w_t) \geq 0$ is a weight function assigned at time point $t = 2, \dots, n$, incorporating model performance. If $g(w_t) = 1, \forall t$, no weighting is applied.
- $F_t^{(j)}$ is the EWMA of past feature importance scores, recursively defined as:

$$\hat{F}_t^{(j)} = \lambda \hat{F}_{t-1}^{(j)} + (1 - \lambda) F_t^{(j)}, \quad (4)$$

where $\lambda \in [0, 1]$ is the smoothing factor controlling the influence of past values.

The process is initialized as $\hat{F}_1^{(j)} = F_1^{(j)}$.

The well-known EWMA is a time series analysis technique that assigns exponentially decreasing weights to past observations. This property renders the EWMA particularly useful for detecting trends and anomalies in noisy data. In contrast to the simple moving average, which assigns equal weight to all past observations, the EWMA assigns greater weight to more recent data points. This characteristic renders the EWMA more responsive to changes in the time series. The parameter λ is a critical component in the analysis, as it governs the rate at which past observations decay. When λ is set to a larger value, it places more emphasis on the most recent data points. Conversely, when λ is set to a smaller value, it gives greater weight to older observations.

The squared difference between the feature importance value and the EWMA of past feature importance scores captures the variability in feature importance over time. The denominator normalizes for the scale of importance values, ensuring stability assessment remains unbiased across features with different magnitudes.

Our approach prioritizes newer models, as they leverage more data, making them a more representative choice than standard mean-based methods. Additionally, the weight function adjusts for model performance, assigning greater

importance to better-performing models while diminishing the influence of models with lower accuracy.

The stability metric is generalizable and can be applied with any feature importance method in Eq. (3). In our experiments, we selected two widely used techniques: Shapley values, which provide an attribution-based measure of feature importance, and tree-based feature importance, derived from decision tree models [6]. The latter is computed by averaging the decrease in impurity (e.g., Gini impurity or entropy) across all splits in which a feature is used. Future work will extend our analysis to additional feature importance metrics, further exploring their impact on stability evaluation.

5 Numerical Results

We start by presenting an illustrative example for the case study in mental health monitoring. Next, experimental validation is provided for a simulated dataset derived from the aforementioned real-world data and two publicly available benchmark time series datasets [3, 19]. Experiments of this work can be found in the publicly available repository [1].

5.1 Illustrative Example for the Real-world Mental Health Monitoring Dataset

The real-world dataset comprises acoustic and psychiatric data collected from a patient diagnosed with bipolar disorder. For further details see [25] with the protocol of this clinical study. Acoustic features describe the manner of speaking with physical descriptors such as shimmer, jitter, and energy extracted from patients’ speech. Additionally, each feature vector is associated with the patient’s mental state at the time of the recording. Bipolar disorder is a serious mental illness characterized by fluctuations from depressive through euthymic to manic states. Previous research confirms that acoustic features extracted from speech serve as valid markers for assessing the severity of manic and depressive symptoms [14]. In this work, we aim to assess the stability of the feature’s importance in time.

To streamline the classification process, the original multiclass classification problem is reformulated into a binary classification by mapping states into two categories: euthymia, which is considered the healthy state, and non-euthymia.

After the feature selection method, four out of 90 features are selected for illustrative purposes of this experiment. There are in total 79 observations available. Observations were split into four different intervals representing different time points.

Fig. 1 depicts the distribution of the two selected features (energy and jitter) showing the difficult nature of solving this problem.

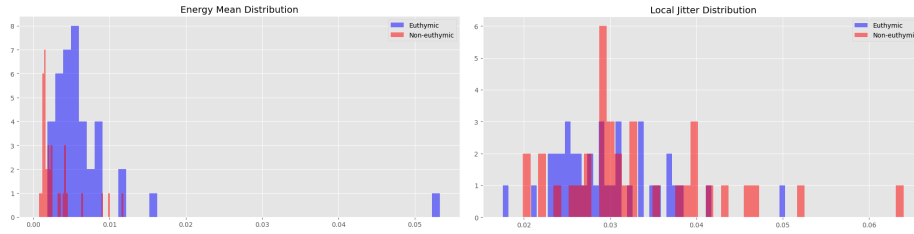


Fig. 1: Distributions of energy and jitter characteristics for real-world data.

Each observation corresponds to a specific day, and the dataset was divided into two categories of periods: ground truth periods, which spanned eight days around a psychiatrist appointment where the patient’s mental state was known, and inter-appointment periods, representing the intervals between psychiatrist visits where the patient’s mental state was inferred. Each ground truth period consisted of 8 observations and each inter-appointment period 11 or 12 observations.

Next, we trained four XGBoost classifiers [7] from the XGBoost package in a sequential manner. The first model was trained on the first ground truth period (8 observations) and tested on the subsequent inter-appointment period (11 observations). The second model was trained on the first two ground truth periods combined with the first inter-appointment period (a total of 27 observations), while the test set consisted of the second inter-appointment period (12 observations). The third and fourth models followed the same pattern, each incorporating more prior data.

For each window, model performance was evaluated using the area under the receiver operating characteristic (ROC) curve (AUC) from the Scikit-learn package. Shapley values, computed with the SHAP package, and feature importance scores from Scikit-learn’s tree-based models were used to analyze the temporal stability of feature attributions. The models achieved an AUC-ROC of 0.50, 0.86, 0.86 and 0.75 for time points 1, 2, 3 and 4, respectively.

We now investigate in detail the stability metric for the test set. Considering the limited length of the dataset, its length is 4 time points. The Shapley values over time for the real-world medical dataset are presented in Fig. 2, while the corresponding stability metric values over time are shown in Fig. 3 for both Shapley values and tree-based feature importance with model performance-based weighting. In these figures, each time point corresponds to the end of each window, i.e., the first time point corresponds to the first ground truth period and inter-appointment period. For each time point, the Shapley values were calculated in the form of a mean based on the corresponding test set, consisting of observations from the inter-appointment period. Averaging Shapley values across all test set observations helps mitigate variability in individual attributions, providing a more stable and interpretable representation of feature importance over time.

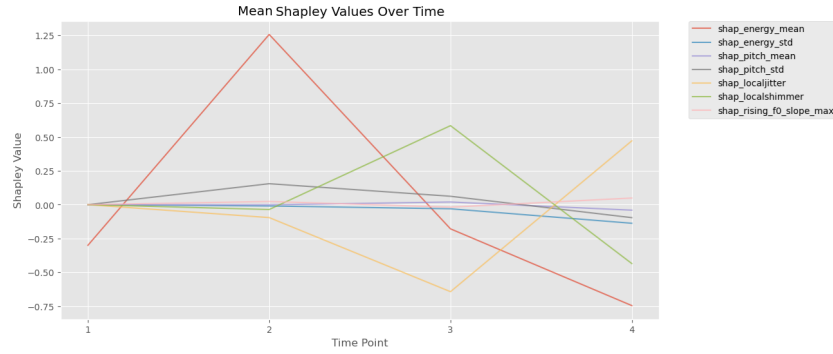


Fig. 2: Changes over time in mean Shapley values for voice features in real-world dataset scenario.

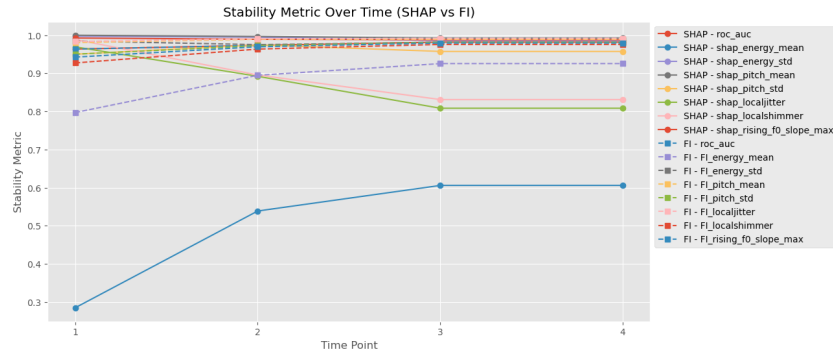


Fig. 3: Changes over time in stability metric for real-world dataset scenario using Shapley values and tree-based feature importance as feature importance metric in Eq. (3).

As observed, although mean energy exhibits the highest absolute Shapley values, its contribution fluctuates significantly across different time points. Consequently, this feature has the lowest stability metric. In contrast, some features with high stability metrics do not necessarily have high Shapley values at every time step. This highlights an important distinction: a high stability metric does not imply a strong influence on model predictions but rather indicates whether a feature’s impact remains consistent over time. Furthermore, the stability metric values for tree-based feature importance are higher than those for Shapley values. This is because tree-based feature importance values in the models were generally lower. However, we observe that the stability metric for Shapley values and tree-based feature importance follows a similar trend for mean energy: both increase and stabilize over time, despite their values differing significantly.

5.2 Simulation study

Considering the limited size of the real data and its complex nature, we simulated a larger dataset that enabled us to validate the proposed metric in a more advanced setting. The resulting dataset is publicly available for further investigation [1]. We selected four voice characteristics—jitter, shimmer, energy, and voice pitch—and used their means and standard deviations from the original dataset to model two distributions: a base distribution representing prior knowledge of mental state classification and a patient-specific distribution incorporating slight variations to reflect individual voice characteristics. Fig. 4 illustrates distributions for two exemplary simulated variables. More figures and details can be found in the publicly available repository [1].

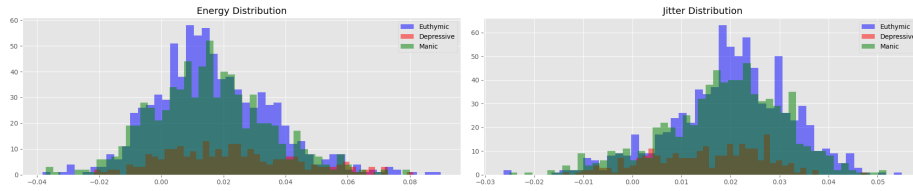


Fig. 4: Distributions of the voice characteristics (energy and jitter) for data simulated for five years.

We then simulated observations over five-year period, assuming a nine-day ground truth period around psychiatrist appointments and a 64-day follow-up period, resulting in five meetings per year. The simulated datasets contained 180 observations, with 60 from each state in the prior knowledge dataset, while the five-year dataset included 1 825 observations with varying label distributions.

We trained XGBoost models using the same methodology as for the real-world dataset, with the addition of prior knowledge from the first step. Model performance was evaluated using ROC AUC or $F1$ -score when ROC AUC was unavailable, and feature attributions were computed via Shapley values and tree-based feature importance using the same packages as in the real-world dataset.

For the simulated medical dataset, Fig. 5 presents Shapley values over time for five-year simulations, along with stability metrics incorporating model weighting by performance. Most stability metric values exceed 0.8, suggesting either minimal fluctuations in Shapley values or that the corresponding features have little influence on the model’s predictions.

However, a closer inspection reveals that jitter, despite having the highest absolute Shapley values, exhibits significant variability in both magnitude and direction over time. This instability is captured in the stability metric, where jitter consistently shows the lowest stability values across both simulation periods.

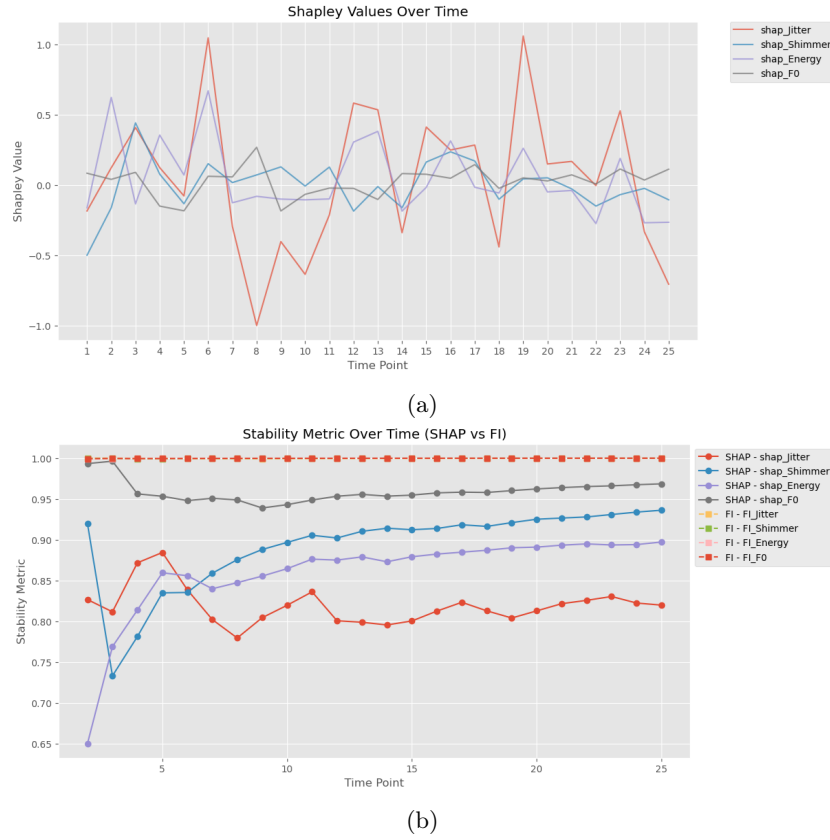


Fig. 5: (a) Changes over time in mean Shapley values for voice features in simulated dataset over five years and (b) stability metric over time for the simulated dataset using Shapley values and tree-based feature importance as feature importance metric in Eq. (3).

Fig. 6 further illustrates changes in Shapley values for jitter and energy, alongside model performance at each time point. Notably, after an initial period of high fluctuation, energy’s Shapley values mostly stabilize at low negative values. However, occasional positive values keep its stability metric at approximately 0.9. In contrast, jitter exhibits persistent fluctuations in both magnitude and direction, resulting in a lower stability metric of around 0.8. The performance-based weighting smooths some Shapley value changes, particularly between time points 21, 22, and 23 for energy. Meanwhile, feature F0 maintains a high stability metric due to its consistently low Shapley values relative to other features. Additionally, the stability metrics for tree-based feature importance remain exceptionally high across all features, reflecting the small absolute values and minimal variations in feature importance observed across models.

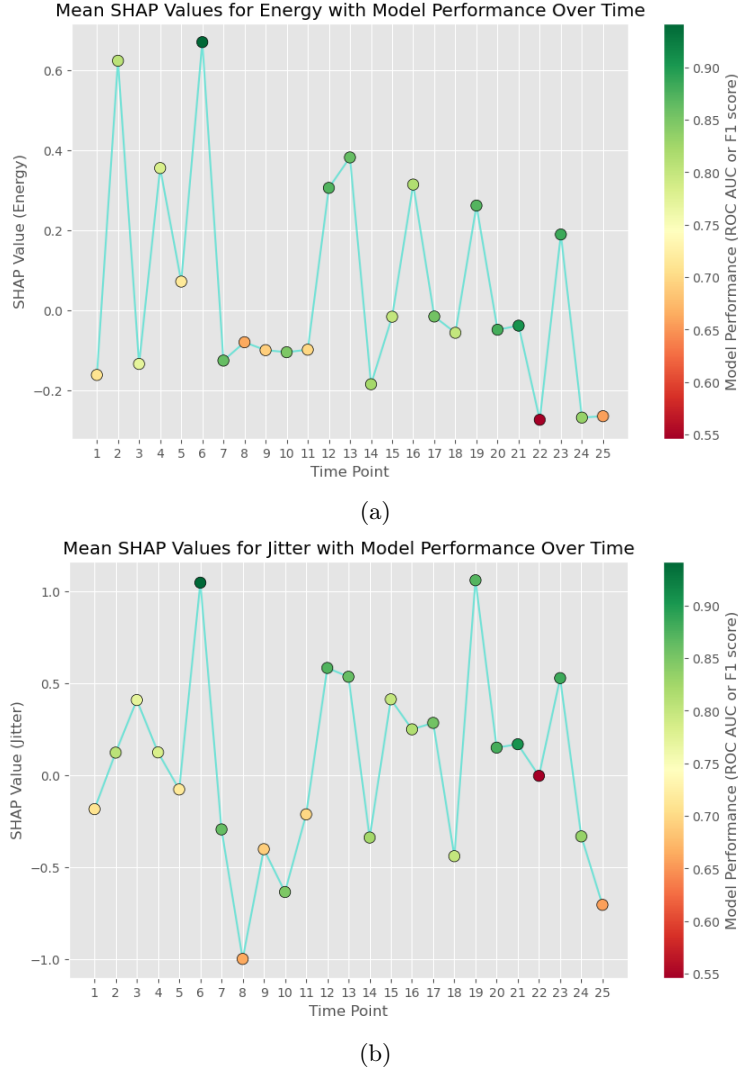


Fig. 6: (a) Changes over time in mean Shapley values for voice features in simulated dataset over five years and (b) stability metric over time for the simulated dataset using Shapley values and tree-based feature importance as feature importance metric in Eq. (3).

5.3 Experiments for benchmarks

To ensure the robustness of our findings, we extended our experimental framework to include publicly available time-series datasets, namely the Predictive Maintenance dataset [3] and Rocket League dataset[19]. The predictive maintenance dataset contains 10000 observations with six explanatory features and

was split into eight intervals, each covering a 30-minute time window. The second dataset consisted of 7189 observations with 16 explanatory features and was divided into 44 intervals of varying lengths, each representing a segment of a Rocket League game. Both datasets were designed for binary classification.

We applied the same window-based data partitioning strategy, XGBoost models, the evaluation metrics and feature importance analyses used for the medical datasets for the datasets described above. Figs. 7 and 9 illustrate the changes in Shapley values for features over time, along with the stability metric values, in the Predictive Maintenance and Rocket League datasets, respectively.

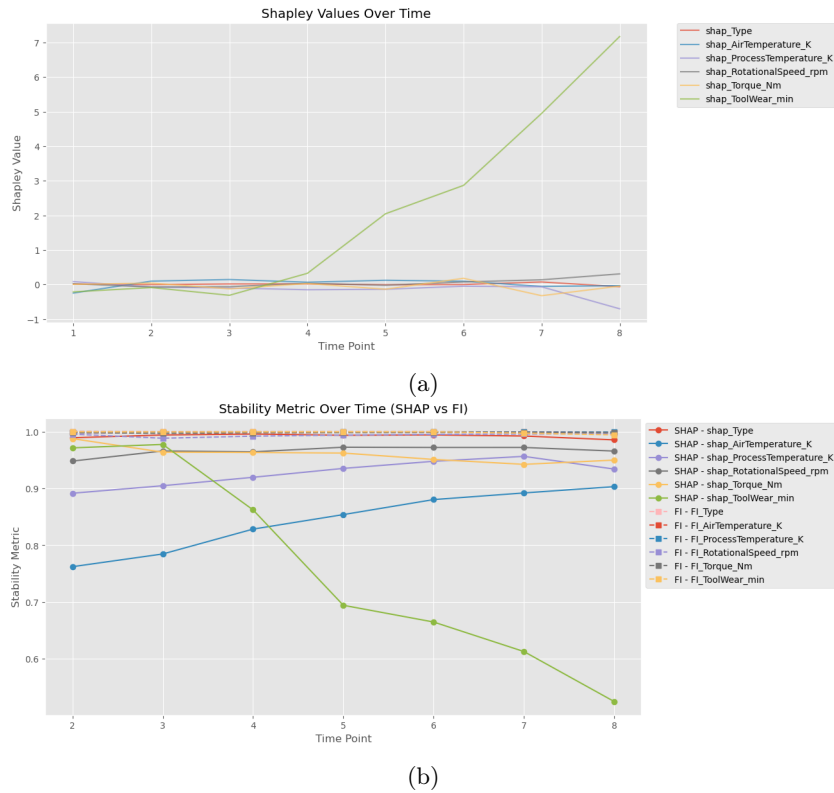


Fig. 7: (a) Changes over time in mean Shapley values for features in Predictive Maintenance dataset and (b) values of stability metric for Predictive Maintenance dataset, with weighting by performance in Eq. (3).

For the Predictive Maintenance dataset, the ToolWear_min feature shows a significant increase in Shapley value over time compared to other features. Consequently, its stability metric decreases, reflecting this consistent upward trend, while the stability metrics for other features stabilize over time.

Furthermore, due to these substantial changes, we observe negative stability metric values for sufficiently high values of λ in Eq. (3), as illustrated in Fig. 8.

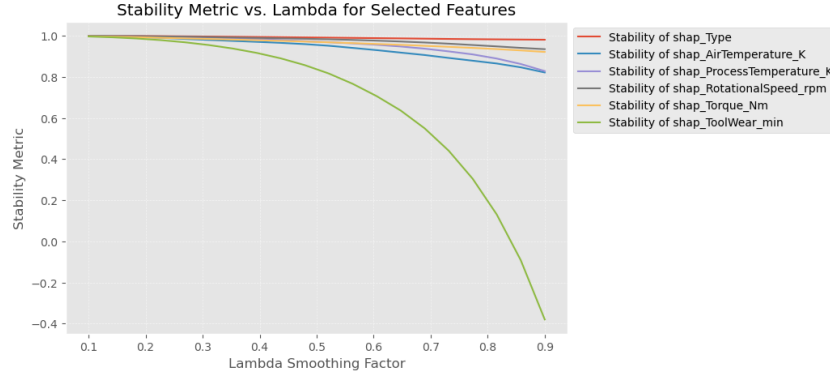


Fig. 8: Changes in values of stability metric for different features in Predictive Maintenance dataset for different values of λ parameter in Eq. (3).

This suggests that under certain conditions—such as irregular yet substantial changes in one direction combined with a sufficiently large λ —the stability metric can become negative. We can observe that stability metric for tree-based feature importance is close to 1 for all values across all time points. This is due to small values and fluctuations taken by the tree-based feature importance for features in this dataset.

As observed in Fig.9, for the Rocket League dataset, the DistanceBall feature starts with the lowest stability metric but stabilizes over time. In contrast, the DistanceCeil and accelerate features show a decreasing stability metric, indicating consistent and significant changes over time. For tree-based feature importance we could observe much smaller values and fluctuations compared to Shapley values, which lead to significantly higher stability metric values.

6 Conclusion and Future Work

In this study, a novel metric for evaluating the stability of feature importance was proposed. In experiments, we considered Shapley local explanations and tree-based feature importance. The proposed metric was developed to capture the nature of temporal data while taking into account model performance. We validated the proposed approach using various datasets, including a real-world medical dataset and two benchmark datasets.

The findings of the proposed approach are encouraging. We conclude that when interpreting model predictions, both the magnitude of a feature’s impact and its stability over time are essential. This reinforces the core motivation behind our approach—providing a more nuanced understanding of feature im-

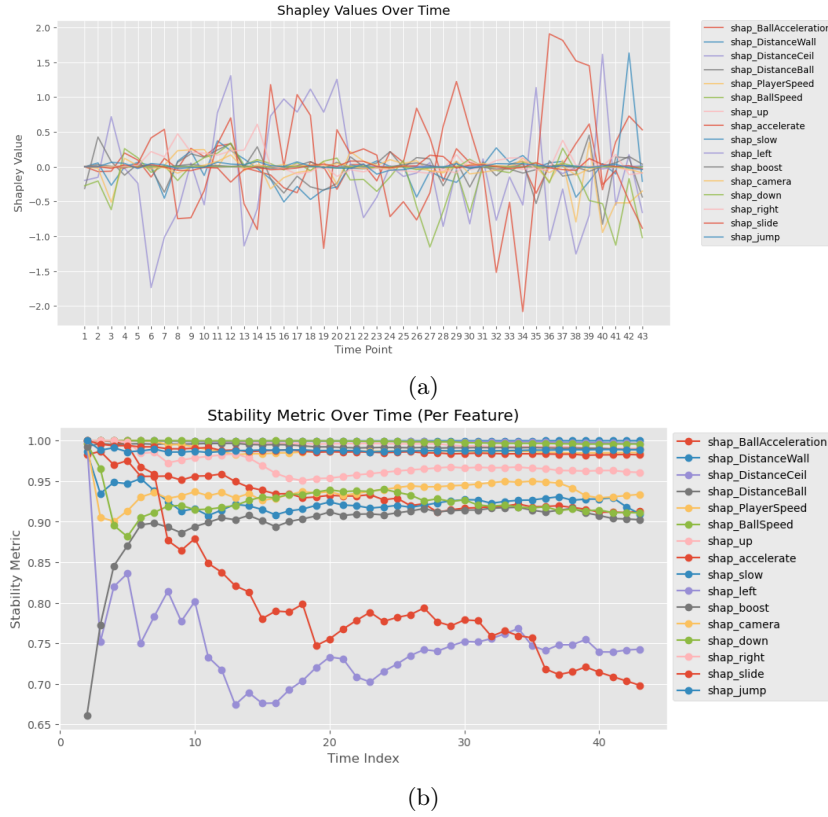


Fig. 9: (a) Changes over time in mean Shapley values for features in Rocket League dataset and (b) values of stability metric for Rocket League dataset, with weighting by performance in Eq. (3).

portance by capturing not only its absolute contribution but also its temporal consistency. While the proposed stability metric offers valuable insights into feature behavior and its influence on model performance, it is most effective when combined with complementary methods. For instance, autocorrelation can reveal how a feature's value evolves over time, providing additional context. By integrating our approach with such methods, one can gain a more comprehensive view of feature dynamics.

Subsequent developments will entail the refinement of the metric through the improvement of weight functions and the incorporation of additional approaches to feature stability, such as autocorrelation. It is hypothesized that these refinements may result in a more intuitive form of the metric, leading to broader applicability and usefulness. Future work will also focus on incorporating additional machine learning and statistical models to further enhance the utility of the proposed approach. We also aim to conduct next experiments using various feature importance metrics on datasets from diverse domains.

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